Stage 7 — Understanding Questions and Answers

Q1. Your two predictions

The ensemble was predicted to score 0.7575 and actually scored 0.7552, so its prediction error was 0.7552 – 0.7575 = –**0.0023**. XGBoost was predicted to score 0.7569 and actually scored 0.7504, giving 0.7504 – 0.7569 = –**0.0065**.

Model	Predicted	Public	Error
Ensemble	0.7575	0.7552	–0.0023
XGBoost	0.7569	0.7504	–0.0065

Predicted ensemble advantage: 0.7575 – 0.7569 = **0.0006**. Actual ensemble advantage: 0.7552 – 0.7504 = **0.0048**, eight times the predicted gap. The ensemble error (0.0023 in absolute value) was within ±0.005; the XGBoost error (0.0065) was outside it.

A small cross-validation advantage is not guaranteed to carry over to a new centre. XGBoost beat elastic-net on 14 of 15 centre-grouped folds but scored 0.7504 on KYW, below elastic-net's 0.7543. KYW is effectively another unseen-centre test, and its patient mix and measurement patterns can differ from the training folds. Combining elastic-net with XGBoost likely reduced dependence on errors specific to either model; this is a plausible robustness explanation, not proof of the precise mechanism.

Q2. A rejected hypothesis

Among the **363 IPSS-R very high-risk patients**, XGBoost achieved an out-of-fold IPCW C-index of **0.6231**, compared with **0.6084** for elastic-net. The improvement was 0.6231 – 0.6084 = **+0.0147**, and XGBoost won **11 of 15 folds**. Thus, the claim that trees generally rank very sick patients worse is not supported by this subgroup test.

This does not eliminate the possibility of extrapolation failure. Validation folds draw patients from the same collection of training centres, so they cannot fully test combinations of disease severity, molecular features, or centre-specific measurements absent from that training collection. KYW has a different case mix; for example, TP53 multi-hit was reported in 18.9% of test patients versus 6.7% of training patients, and complex karyotypes in 31.3% versus 10.9%. The evidence rejects the simple "trees are bad for very sick patients" explanation, while leaving the cause of the external performance gap unresolved.

Q3. The main lesson

Improvement	Mean CV C-index gain
Benchmark Cox → IPSS-M-style Cox (24 features)	+0.0278
IPSS-M-style Cox → elastic-net (84 features)	+0.0059

Improvement	Mean CV C-index gain
Elastic-net → two-model ensemble	+0.0032

The largest improvement came from the clinically informed move from the benchmark to the 24-feature IPSS-M-style model. Its gain of 0.0278 was approximately 4.7 times the 24-to-84-feature gain and 8.7 times the ensemble gain.

Job interview answer: This project showed me that medically informed feature engineering can matter more than increasing algorithmic complexity. Adding clinically meaningful information such as TP53 multi-hit status and mutation burden substantially improved survival-risk ranking, whereas expanding the feature set and ensembling added smaller gains. I learned to evaluate models using centre-grouped validation rather than relying on random splits, because an unseen hospital can have a different case mix. Finally, the public results taught me to favour stable generalization over a small cross-validation advantage: our ensemble scored 0.7552, while XGBoost alone scored 0.7504.